Program Evaluation with Observational Data

Chapter 6 - Regression Discontinuity Design

Advanced Empirical Research Method

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Introduction

So far, our solutions to the fundamental problem of causal inference:

1. **Selection on observable:** similar individuals/units are defined by their observable characteristics.
2. **Difference-in-difference:** similar individuals/units shared the same trend as the treated group before the program implementation.

Introduction

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1. **Selection on observable:** similar individuals/units are defined by their observable characteristics.
2. **Difference-in-difference:** similar individuals/units shared the same trend as the treated group before the program implementation.

Today:

1. **Regression Discontinuity Design:** Similar individuals/units are those very close to a defined cut-off, but have not been taken in the program.

Introduction

Often access to, or incentives for participation in, a service or program is assigned based on transparent rules with criteria based on **clear cutoff** values, rather than on discretion of administrators.

Comparisons of individuals that are **similar but on different sides** of the cutoff point can be credible estimates of causal effects for a specific subpopulation.

Discontinuity Design

Two potential outcomes $Y_i(0)$ and $Y_i(1)$, defining a causal effect $Y_i(1) - Y_i(0)$.

Define X_i the score (running variable), and c the cutoff. The treatment assignment rule is :

$$D_i = I\{X_i \geq c\}$$

Therefore:

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i) = \begin{cases} Y_i(0) & \text{if } X_i < c \\ Y_i(1) & \text{if } X \geq c \end{cases}$$

The fundamental problem of causal inference occurs because we only observe the outcome under control, $Y_i(0)$, for units whose score is below c, and vice-versa.

Sharp RDD: Illustration

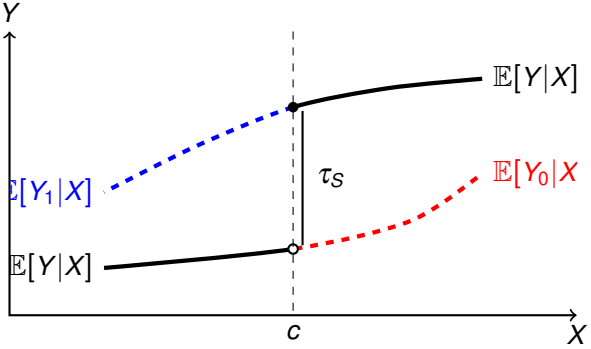


Figure: Outcomes in the sharp regression discontinuity design. For $X < c$ we observe $\mathbb{E}[Y|X] = \mathbb{E}[Y_0|X]$; for $X \geq c$ we observe $\mathbb{E}[Y|X] = \mathbb{E}[Y_1|X]$. The "jump" in $\mathbb{E}[Y|X]$ at c is the conditional ATE when $X = c$, namely $\tau_S \equiv \mathbb{E}[Y_1 - Y_0|X = c]$.

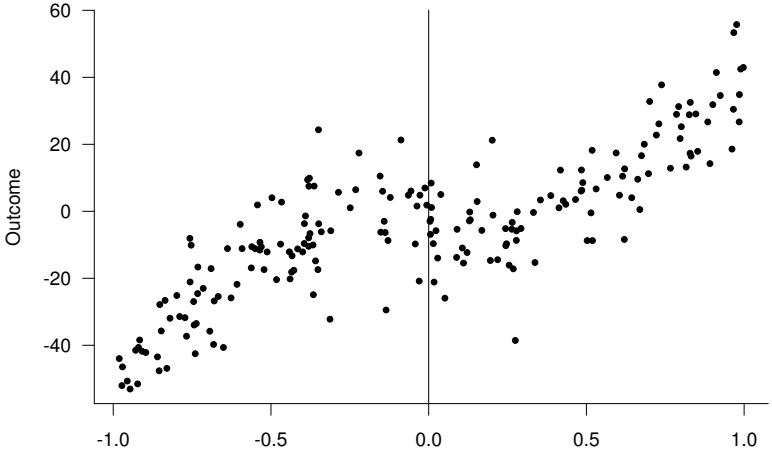
Discontinuity Design

The ATE for any given X is the vertical distance between the two regression curves at the value of the running variable.

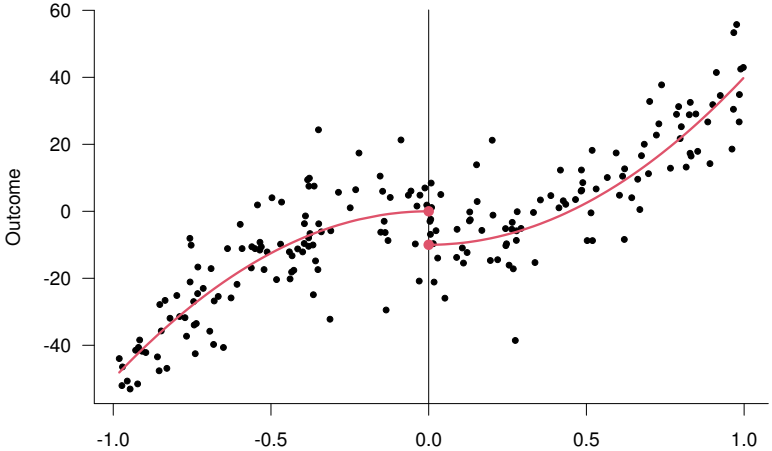
This distance **cannot** be directly estimated because we **never observe** both curves for the same value of x.

However, a special situation occurs at the cutoff c: this is the only point at which we 'almost' observe both curves.

RD Design



RD Design



Sharp RDD

Sharp Regression Discontinuity Design:
Everyone on one side of the cutoff is treated and everyone on the other side is not, so the treatment variable "jumps" from zero to one at the cutoff.

$$D_i = I\{X_i \geq c\}$$

Basic idea:
Compare those who are slightly above the cutoff to those who are slightly below.

- Assume continuity of outcome at cutoff, i.e. no manipulation.
- compare mean outcomes for data near the threshold.

Fuzzy RDD

Fuzzy Regression Discontinuity Design:
Everyone on one side of the cutoff is **more likely** treated than everyone on the other side, so the *probability of being treated* "jumps" at the cutoff.

$$\lim_{x \downarrow c} \mathbb{P}(D = 1 | X = x) \neq \lim_{x \uparrow c} \mathbb{P}(D = 1 | X = x)$$

Basic idea:
Compare those who are slightly above the cutoff to those who are slightly below.

- Assumes continuity of outcome at cut-off, i.e. no manipulation;
- compare mean outcomes for data near the threshold;
- "Magnify" this effect by the size of the jump: Wald estimand.

Sharp or Fuzzy RDD: Montizaan, Cörvers, and De Grip (2010)

Sometimes, what you consider as a Sharp of Fuzzy RDD depends on how you define the treatment.

Major pension reform if the Dutch public sector in 2006.

If employee were **born before 1949** and had been working in the public sector continuously since April 1, 1997, they could still **retire at age 62 and 3 months**, with a pension offering 70 percent replacement rate.

Employee born after 1950 were subject to less generous benefits: you could reach the same replacement rate as before by working one additional year.

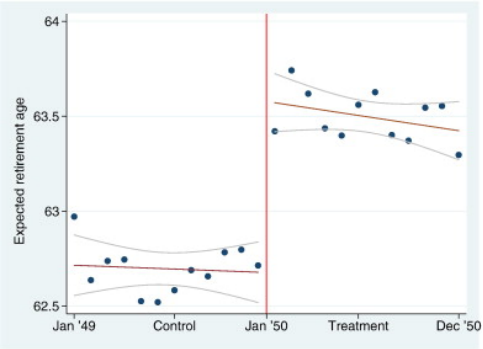
Sharp or Fuzzy RDD: Montizaan, Cörvers, and De Grip (2010)

If the treatment is the ‘eligibility to early retirment’ among public sector worker, then you have a sharp RDD. Anyone born in 1950 and after is affected by the treatment.

If the treatment is the ‘take-up of early retirement’ among public workers, then you have a fuzzy RDD. Only a proportion of the population is induced to differ their retirement.

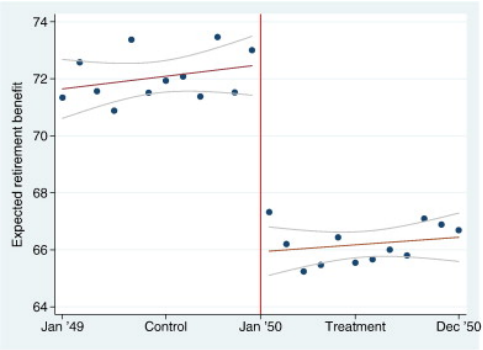
RDD example: Montizaan, Cörvers, and De Grip (2010)

Figure: Effect on retirement age



RDD example: Montizaan, Cörvers, and De Grip (2010)

Figure: Effect on Pension benefits



Sharp RDD: Assumptions

Assumption (Sharp RD Design)

$D = I\{X \geq c\}$ where X is an observed covariate, and c is a known threshold.

Interpretation:
Treatment is completely determined by the running variable X and a cutoff c that is known to the researcher.

Note: There is in a strict sense a **selection on observable** characteristics (the running variable), but there is no overlap!

Sharp RDD: Assumptions

Assumption (Continuity of Conditional Means)

$\mathbb{E}[Y_0|X = x]$ and $\mathbb{E}[Y_1|X = x]$ are both continuous functions of x at the point $x = c$.

Interpretation:
Individuals whose running variable falls in a sufficiently small neighbourhood around the treatment cutoff are "effectively identical."
Rules out a "jump" in Y_0 or Y_1 as X moves from just below to just above the cut-off c.

Sharp RDD: Identification result

Theorem (Sharp Regression Discontinuity)

Under Assumptions Sharp RDD and Continuity of Conditonal Mean,

$$\tau_S \equiv \mathbb{E}[Y_1 - Y_0 | X = c] = \lim_{x \downarrow c} \mathbb{E}[Y | X = x] - \lim_{x \uparrow c} \mathbb{E}[Y | X = x].$$

Interpretation:

τ_S: causal effect of treatment for someone whose score is *close to the treatment cutoff*.

A well-defined causal effect, but may differ from the overall average treatment effect.

Falsification Tests

The key identifying assumption in the RDD setup is:
$E(Y_i(0)|X_i)$ and $E(Y_i(1)|X_i)$ are continuous in X_i at $X_i = c$.

This means that all other unobserved determinants of Y are continuously related to the running variable X at $X = c$.

This assumption allows the researcher to use average outcomes of units just below the cutoff as a valid counterfactual for units right above the cutoff.

This assumption cannot be directly tested. But there are some tests which give suggestive evidence of whether or not the assumption is valid.

Falsification Tests

Use estimators from two separate samples, below and above the cutoff value:

1. Plot regression function $E[Y_i|X_i = x]$
2. Plot regression functions $E[Z_i|Xi = x]$ for covariates that do not enter the assignment rule D_i.
3. Plot density $f_X(x)$.

In addition, one should also inspect the graph to see whether there are any **other jumps in the conditional expectation** of Y given X that are comparable to, or larger than, the discontinuity at the cutoff value. If so, and if one cannot explain such jumps on substantive grounds, it would call into question the interpretation of the jump at the threshold as the causal effect of the treatment.

Falsification Tests: regression functions

For some bandwidth h and number of bins K_l, K_r, for $k = 1, \ldots, K$ with $K = K_l + K_r$, define the bins $[b_{k-1}, b_k]$ where $b_k = c - (K_0 - k + 1)h$. Calculate the average outcome and average covariates in each bin:

$$\begin{aligned} \overline{Y}_k &= \frac{1}{N_k} \sum_{i=1}^{N} Y_i I\{b_k \leq X_i < b_k + 1\}. \\ \overline{Z}_k &= \frac{1}{N_k} \sum_{i=1}^{N} Z_i I\{b_k \leq X_i < b_k + 1\}. \end{aligned}$$

In practice, the **visual clarity** of the plots is often improved by adding smoothed regression lines based on polynomial regressions (or other flexible methods) estimated on the raw data.

Falsification Tests: Density of the running variable

An approximation of the density:

$$N_k = \sum_{i=1}^{N} I\{b_k \leq X_i < b_k + 1\}.$$

This plot can be used to inspect whether there is a discontinuity in the distribution of the running variable X at the threshold. Such discontinuity would raise the question of whether the value of this covariate was manipulated by the individual agent, invalidating the design (see Mc Crary, 2008).

For example, suppose that the running variable is a test score. If individuals know the threshold and have the option of **retaking the test**, individuals with test scores just below the threshold may do so, and invalidate the design.

Sharp RDD: Regression Model

Because there is no value of X_i at which you observe both treatment and control observations, the method relies on extrapolation across covariate values: we **cannot be agnostic** about regression functional form in RD.

Sharp RDD: Parametric Linear Regression Model

Linear framework with homogeneous covariate effects:

$$Y_i(d) = \alpha(d) + \beta X_i + U_i(d), \text{ with } E\left(U_i(d)|X_i, D_i = d_i\right) = 0$$

Then one gets the regression:

$$Y_i = \alpha(0) + \beta X_i + D_i(\alpha(1) - \alpha(0)) + \varepsilon_i, \text{ with } E\left(\varepsilon_i|X_i, D_i\right) = 0$$

The key difference between this regression and regressions in previous lectures is that D_i is not only correlated with X_i but it is a **deterministic function** of X_i.

The treatment parameter: $\tau = \alpha(1) - \alpha(0)$.

Sharp RDD: Parametric Non-Linear Regression Model

Non-Linear framework :

$$Y_i(d) = \alpha(d) + \sum_{j=1}^{p} \beta(d)\,(X_i - c)^j + U_i(d), \text{ with } E\left(U_i(d)|X_i, D_i = d_i\right) = 0$$

We have include 'power terms': this is usually called higher order polynomials.

Don't use global higher order polynomials [Gelman and Imbens, 2019]:

(i) Runge's Phenomenon - estimation is bad at boundaries,

(ii) Polynomial order is arbitrary,

(iii) Inference based on high-order polynomials is often poor.

Sharp RDD: Non-parametric Regression Models

The standard Kernel regression below generates bias at the boundary:

$$\widehat{\mu_l(c)} = \frac{\sum_{i=1}^{N} Y_i K_d^{-}\left((c - X_i)/h_n\right)}{\sum_{i=1}^{N} K_d^{-}\left((c - X_i)/h_n\right)} \text{ and } \widehat{\mu_r(c)} = \frac{\sum_{i=1}^{N} Y_i K_d^{+}\left((c - X_i)/h_n\right)}{\sum_{i=1}^{N} K_d^{+}\left((c - X_i)/h_n\right)}$$

where:

$$\begin{aligned} K_d^{-}\left((c - X_i)/h_n\right) &= K_d\left((c - X_i)/h_n\right) I(X_i < c) \text{ and} \\ K_d^{+}\left((c - X_i)/h_n\right) &= K_d\left((c - X_i)/h_n\right) I(X_i \geq c) \end{aligned}$$

For example, and often in pratice, using a uniform kernel:
$K_d^{-}\left((c - X_i)/h_n\right) = I\{c - h \leq X_i < c\}$ and $K_d^{+}\left((c - X_i)/h_n\right) = I\{c \leq X_i \leq c + h\}$.

Sharp RDD: Non-parametric Regression Models

The local linear regression does not have the boundary problem:

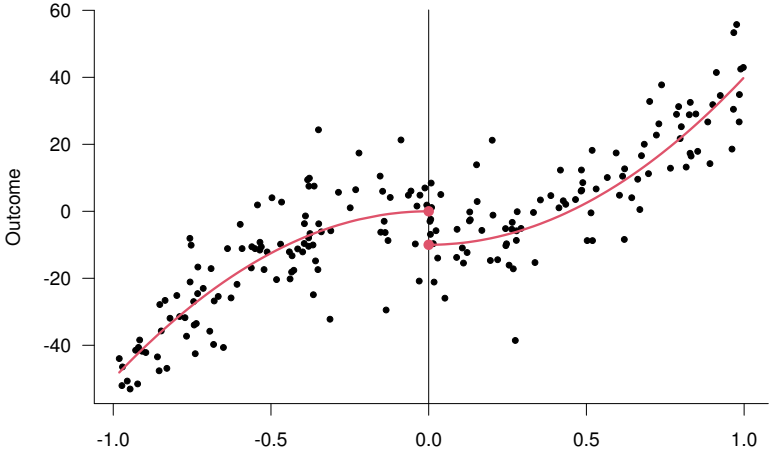
$$\text{argmin}_{\alpha_l,\beta_l} \sum_{i=1}^{n} \left(y_i - \alpha_l - \beta_l(x_i - c)\right)^2 K_d^{-}\left(\left(c - X_i\right)/h_n\right) \text{ and}$$
$$\text{argmin}_{\alpha_r,\beta_r} \sum_{i=1}^{n} \left(y_i - \alpha_r - \beta_r(x_i - c)\right)^2 K_d^{+}\left(\left(c - X_i\right)/h_n\right).$$

Note that: $\widehat{\mu_l(c)} = \hat{\alpha}_l + \hat{\beta}_l(c - c) = \hat{\alpha}_l$, and $\widehat{\mu_r(c)} = \hat{\alpha}_r$. Then:

$$\widehat{\tau}_Y(h) = \hat{\alpha}_l - \hat{\alpha}_r$$

.

Local Linear Regression



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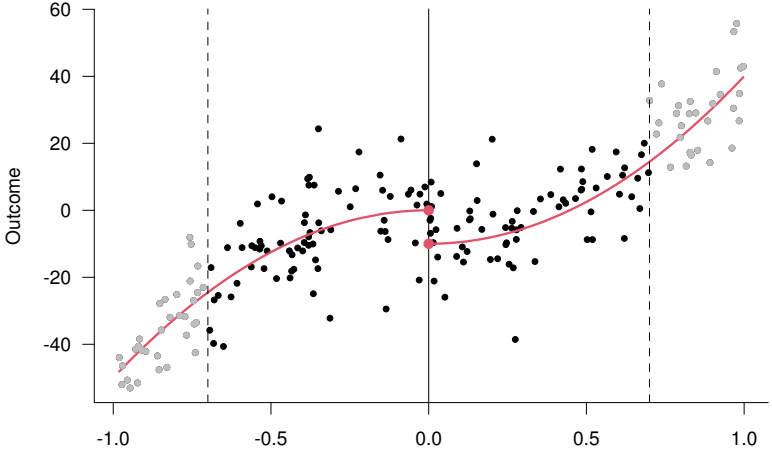

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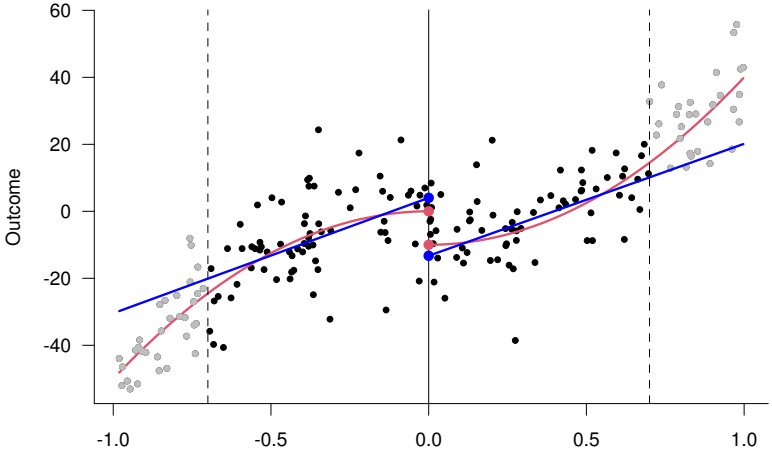

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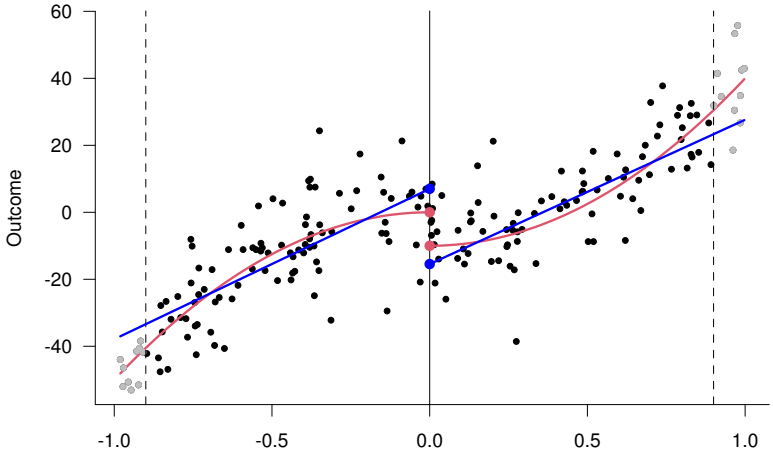
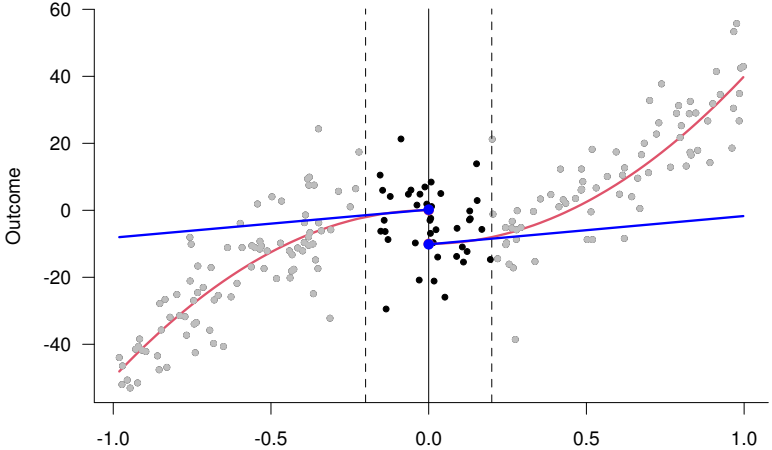


Local Linear Regression



Sharp RDD: Non-parametric Regression Model

Alternatively one can estimate the average effect directly in a single local linear regression:

$$\underset{\alpha,\beta,\tau,\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \alpha - \beta_l(x_i - c) - \tau D_i - \gamma(X_i - c)D_i\right)^2 K_d\left((c - X_i)/h_n\right)$$

which will numerically yield the same estimate of $\hat{\tau}$.

Two key problems:

1. How to choose the bandwidth h?
2. How to construct valid confidence sets (CS) for $\widehat{\tau}_Y(h)$.

Choosing the bandwidth

Heuristically, if $h \searrow$, then $\text{bias}(\widehat{\tau}_Y(h)) \searrow$ and $\text{se}(\widehat{\tau}_Y(h)) \nearrow$.

- The choice of bandwidth trades-off bias and variance of $\widehat{\tau}_Y(h)$, e.g. minimize MSE

$$\text{Bias}(\widehat{\tau}_Y(h))^2 + \text{Var}(\widehat{\tau}_Y(h))$$

- Imbens and Kalyanaraman (2012) provide a formula for "MSE-optimal" bandwidth. It is of the form:

$$h_{MSE} = C_{MSE}^{1/5} \times n^{-1/5} \qquad C_{MSE} = C(K) \times \frac{\text{Var}(\widehat{\tau}_Y(h))}{\text{Bias}(\widehat{\tau}_Y(h))^2}$$

- A "Pointwise" feasible h_{MSE} is obtained by plugging-in estimates of $\text{Var}(\widehat{\tau}_Y(h))$ and $\text{Bias}(\widehat{\tau}_Y(h))$, but this procedure can be quite bad in practice.

Choosing the bandwidth

An alternative is the leave-out-one cross-validation.

- Check how well a regression of Y on X predicts the value of Y in the data, when trying a range of bandwidth values.
- When predicting Y_i with badwidth h, we leave out i out of the sample.
- To mimic RDD, we do a regression on the left and on the right.
- Ludwig and Miller (2007) suggest to pick the bandwidth that minimises the MSE.

Kettlewell and Siminski (2020) is another alternative: Compute placebos and pick the bandwidth that bring the placebos closest to zero.

Confidence sets for jump τ_Y

Under fairly general conditions, it holds that

$$\frac{\widehat{\tau}_Y(h) - \tau_Y}{\mathrm{se}(\widehat{\tau}_Y(h))} = \frac{\widehat{\tau}_Y(h) - \tau_Y - \mathrm{bias}(\widehat{\tau}_Y(h))}{\mathrm{se}(\widehat{\tau}_Y(h))} + \frac{\mathrm{bias}(\widehat{\tau}_Y(h))}{\mathrm{se}(\widehat{\tau}_Y(h))} \overset{d}{\approx} N\left(\frac{\mathrm{bias}(\widehat{\tau}_Y(h))}{\mathrm{se}(\widehat{\tau}_Y(h))}, 1\right).$$

Standard error straightforward to obtain. Main issue is handling of bias.

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Standard error straightforward to obtain. Main issue is handling of bias.

Let $\text{cv}_{1-\alpha}(r)$ be $1-\alpha$ quantile of $|N(r,1)|$, so $\text{cv}_{.95}(0) = 1.96$. Naive approach simply ignores smoothing bias and that leads to CSs that are incorrectly centered in finite samples

$$CI_{naive} = [\widehat{\tau}_Y(h_{MSE}) \pm \text{cv}_{.95}(0) \times \text{se}(\widehat{\tau}_Y(h_{MSE}))].$$

Common Approaches to Sharp RD Inference

- Undersmoothing assumes "small" bandwidth s.t. $\dfrac{\text{bias}(\widehat{\tau}_Y(h))}{\widehat{\text{se}}(\widehat{\tau}_Y(h))} \to 0$:

$$\text{CI}_{US} = [\widehat{\tau}_Y(h_{\text{small}}) \pm \text{cv}_{1-\alpha}(0) \cdot \widehat{\text{se}}(\widehat{\tau}_Y(h_{\text{small}}))].$$

- Robust bias correction adjusts point estimate and standard error (Calonico, Cattaneo, Titiunik, 2014, Calonico, Cattaneo, Farell, 2020):

$$\text{CI}_{RBC} = [\widehat{\tau}_Y^{bc}(h_{\text{mse}}) \pm \text{cv}_{1-\alpha}(0) \cdot \widehat{\text{se}}^{bc}(\widehat{\tau}_Y(h_{\text{mse}}))].$$

R-package: rdrobust.

Common Approaches to Sharp RD Inference

- Bias-aware approach adjusts cv for the "worst-case" bias (Armstrong and Kolesár, 2018):

$$\mathrm{CI}_{BA} = [\widehat{\tau}_Y(h_{\mathrm{opt}}) \pm \mathrm{cv}_{1-\alpha}(\widehat{r}(h_{\mathrm{opt}})) \cdot \widehat{\mathrm{se}}((\widehat{\tau}_Y(h_{\mathrm{opt}})))].$$

  where $\mathrm{cv}_{1-\alpha}(\widehat{r}(h_{\mathrm{opt}})) > 1.96$. R-package: RDHonest.
- Randomisation inference: Cattaneo, Frandsen, and Titiunik (2015).
  (1) Repeatedly randomly permuting observations close to the threshold to be part of the treated or nontreated group;
  (2) Verify how extreme the estimated $\widehat{\tau}$ based on the actual treatment is relative to the permuted effects.

Predetermined Covariates

Predetermined covariates W_i are often include linearly in the regression equation:

$$\underset{\alpha,\beta,\tau,\gamma}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \alpha - \beta_l(x_i - c) - \tau D_i - \gamma(X_i - c) D_i - \underbrace{\gamma' W_i}_{\mu_\gamma(W_i)} \right)^2 K_d\left((c - X_i)/h_n\right)$$

If there are only "few" covariates, $\widehat{\tau}_Y(h, \mu_\gamma)$ is a consistent estimator for τ_Y, but not necessarily efficient.

Noack et al. [2021] propose a new class of RD estimators that is more efficient:

- $\mu_\gamma(W_i)$ is allowed to be basically arbitrarily chosen.
- Bandwidth choice and conducting inference follow common approaches.

Falsification tests

Practice...

Fuzzy RDD

Fuzzy Regression Discontinuity Design:
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Basic idea:
Compare those who are slightly above the cutoff to those who are slightly below.

- Assumes continuity of outcome at cut-off, i.e. no manipulation;
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- "Magnify" this effect by the size of the jump: Wald estimand.

Fuzzy RDD

Assumption (Fuzzy RD Design)

$$\lim_{x \downarrow c} \mathbb{P}(D = 1 | X = x) \neq \lim_{x \uparrow c} \mathbb{P}(D = 1 | X = x)$$

where X is an observed covariate, and c is a known threshold.

Interpretation:
requires a jump in the *probability of treatment* at the cutoff c.
Equivalent to the IV first-stage assumption.

Fuzzy RDD: Estimand

$$\tau_F \equiv \frac{\lim_{x\downarrow c}\mathbb{E}[Y|X=x] - \lim_{x\uparrow c}\mathbb{E}[Y|X=x]}{\lim_{x\downarrow c}\mathbb{E}[D|X=x] - \lim_{x\uparrow c}\mathbb{E}[D|X=x]}. \tag{1}$$

Interpretation:

The Wald estimand "magnifies" the ITT by dividing it by the IV first-stage.

τ_F divides the sharp RD estimand τ_S by the equivalent fuzzy RD "first-stage."

Question

Can we give τ_F a meaningful causal interpretation?

Fuzzy RDD: Causal interpretation

View D as a *deterministic function* of x for each individual.

$$D : x \mapsto D(x) \in \{0, 1\}$$

For any possible value x of X, $D(x)$ treatment compliance, $D(x) = 1$, or not, $D(x) = 0$.

Fuzzy RD Complier: A person for whom $D(x)$ switches from 0 to 1 as x crosses the threshold c.

Fuzzy RDD: Causal interpretation

Assumption (Monotonicity)

There is some $\varepsilon > 0$ such that $D(c-h) \leq D(c+h)$ for all $0 < h < \varepsilon$.

Assumption (Exclusion Restriction)

$(Y_1 - Y_0)$ and $D(x)$ are jointly independent of X near c.

Fuzzy RDD: Causal interpretation

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Assumption (Exclusion Restriction)

$(Y_1 - Y_0)$ and $D(x)$ are jointly independent of X near c.

Interpretation:

Violation of monotonicity: Participants who only receive treatment when their score *disqualifies them*.

Close to the cutoff, both a person's compliance type, as determined by $D(x)$, and her treatment effect $(Y_1 - Y_0)$ are independent of her test score X.

Fuzzy RDD: Causal interpretation

Theorem

Suppose that $\mathbb{E}[Y_0|X = x]$ *is a continuous function of* x *at* $x = c$*. Then, under Assumptions Monotonicity and Exclusion, the fuzzy RD estimand* τ_F *from* (1) *satisfies*

$$\tau_F = \lim_{h \downarrow 0} \mathbb{E}[Y_1 - Y_0 | D(c+h) - D(c-h) = 1].$$

Interpretation:

τ_F represents the average treatment effect for compliers at $X = c$.

Fuzzy RD is really a *sharp* RD for compliers at $X = c$.

Fuzzy RDD: Causal interpretation

$$\mathbb{E}[Y|X] = \mathbb{E}[Y_0 + D(Y_1 - Y_0)|X] = \mathbb{E}[Y_0|X] + \mathbb{E}[D(Y_1 - Y_0)|X]$$

and hence

$$\lim_{x \downarrow c} \mathbb{E}[Y|X = x] = \lim_{x \downarrow c} \mathbb{E}[Y_0|X = x] + \lim_{x \downarrow c} \mathbb{E}[D(Y_1 - Y_0)|X = x]$$
$$\lim_{x \uparrow c} \mathbb{E}[Y|X = x] = \lim_{x \uparrow c} \mathbb{E}[Y_0|X = x] + \lim_{x \uparrow c} \mathbb{E}[D(Y_1 - Y_0)|X = x].$$

Fuzzy RDD: Causal interpretation

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$$\lim_{x \uparrow c} \mathbb{E}[Y|X = x] = \lim_{x \uparrow c} \mathbb{E}[Y_0|X = x] + \lim_{x \uparrow c} \mathbb{E}[D(Y_1 - Y_0)|X = x].$$

Since $\mathbb{E}[Y_0|X = x]$ is continuous at c:

$$\begin{aligned}\lim_{x \downarrow c} \mathbb{E}[Y|X = x] - \lim_{x \uparrow c} \mathbb{E}[Y|X = x] &= \lim_{x \downarrow c} \mathbb{E}[D(Y_1 - Y_0)|X = x] - \lim_{x \uparrow c} \mathbb{E}[D(Y_1 - Y_0)|X = x] \\ &= \lim_{h \downarrow 0} \{\mathbb{E}[D(Y_1 - Y_0)|X = c + h] - \mathbb{E}[D(Y_1 - Y_0)|X = c - h]\}.\end{aligned}$$

Fuzzy RDD: Causal interpretation

Now, for $0 < h < \varepsilon$ Assumption *Exclusion* gives

$$
\begin{aligned}
&\mathbb{E}[D(Y_1 - Y_0)|X = c+h] - \mathbb{E}[D(Y_1 - Y_0)|X = c-h] \\
&\qquad = \mathbb{E}[D(c+h)(Y_1 - Y_0)|X = c+h] - \mathbb{E}[D(c-h)(Y_1 - Y_0)|X = c-h] \\
&\qquad = \mathbb{E}[D(c+h)(Y_1 - Y_0)] - \mathbb{E}[D(c-h)(Y_1 - Y_0)] \\
&\qquad = \mathbb{E}[\{D(c+h) - D(c-h)\}(Y_1 - Y_0)] \\
&\qquad = \mathbb{E}[Y_1 - Y_0|D(c+h) - D(c-h) = 1]\,\mathbb{P}[D(c+h) - D(c-h) = 1] \\
&\qquad = \mathbb{E}[Y_1 - Y_0|D(c+h) - D(c-h) = 1]\,\mathbb{E}[D(c+h) - D(c-h)]
\end{aligned} \tag{2}
$$

since $[D(c+h) - D(c-h)] \in \{0, 1\}$ by Assumption *Monotonicity*.

Fuzzy RDD: Causal interpretation

Now, for $0 < h < \varepsilon$ Assumption *Exclusion* gives

$$
\begin{aligned}
&\mathbb{E}[D(Y_1 - Y_0)|X = c+h] - \mathbb{E}[D(Y_1 - Y_0)|X = c-h] \\
&\qquad = \mathbb{E}\left[D(c+h)(Y_1 - Y_0)|X = c+h\right] - \mathbb{E}\left[D(c-h)(Y_1 - Y_0)|X = c-h\right] \\
&\qquad = \mathbb{E}\left[Y_1 - Y_0|D(c+h) - D(c-h) = 1\right]\mathbb{P}\left[D(c+h) - D(c-h) = 1\right] \\
&\qquad = \mathbb{E}\left[Y_1 - Y_0|D(c+h) - D(c-h) = 1\right]\mathbb{E}\left[D(c+h) - D(c-h)\right]
\end{aligned}
\tag{2}
$$

since $[D(c+h) - D(c-h)] \in \{0, 1\}$ by Assumption *Monotonicity*.

Fuzzy RDD: Causal interpretation

Now, for $0 < h < \varepsilon$ Assumption *Exclusion* gives

$$
\begin{aligned}
&\mathbb{E}[D(Y_1 - Y_0)|X = c+h] - \mathbb{E}[D(Y_1 - Y_0)|X = c-h] \\
&\qquad = \mathbb{E}\left[D(c+h)(Y_1 - Y_0)|X = c+h\right] - \mathbb{E}\left[D(c-h)(Y_1 - Y_0)|X = c-h\right] \\
&\qquad = \mathbb{E}\left[D(c+h)(Y_1 - Y_0)\right] - \mathbb{E}\left[D(c-h)(Y_1 - Y_0)\right] \\
&\qquad = \mathbb{E}\left[\{D(c+h) - D(c-h)\}(Y_1 - Y_0)\right] \\
&\qquad = \mathbb{E}\left[Y_1 - Y_0|D(c+h) - D(c-h) = 1\right]\mathbb{P}\left[D(c+h) - D(c-h) = 1\right] \\
&\qquad = \mathbb{E}\left[Y_1 - Y_0|D(c+h) - D(c-h) = 1\right]\mathbb{E}\left[D(c+h) - D(c-h)\right]
\end{aligned} \tag{2}
$$

since $[D(c+h) - D(c-h)] \in \{0, 1\}$ by Assumption *Monotonicity*.

Fuzzy RDD: Causal interpretation

Similarly,

$$\begin{aligned}\mathbb{E}[D|X=c+h]-\mathbb{E}[D|X=c-h] &= \mathbb{E}[D(c+h)|X=c+h]-\mathbb{E}[D(c-h)|X=c-h] \\ &= \mathbb{E}[D(c+h)-D(c-h)].\end{aligned}$$

The result follows by substituting the preceding into the final equality of (2), since

$$\lim_{x\downarrow c}\mathbb{E}[D|X=x]-\lim_{x\uparrow c}\mathbb{E}[D|X=x]=\lim_{h\downarrow 0}\mathbb{E}[D(c+h)-D(c-h)].$$

Fuzzy RDD: Parametric Regression model

In a linear framework with homogeneous covariate effects, two reduced form regressions:

$$\begin{aligned} D_i &= \alpha_d(0) + X_i\beta_d + I\{X_i > c\}(\alpha_d(1) - \alpha_d(0)) + u_i, \text{ with } E(u_i|X_i) = 0 \\ Y_i &= \alpha_y(0) + X_i\beta_y + I\{X_i > c\}(\alpha_y(1) - \alpha_y(0)) + \varepsilon_i, \text{ with } E(\varepsilon_i|X_i) = 0 \end{aligned}$$

$\alpha_d(1) - \alpha_d(0)$ is akin to the first-stage in an IV regression.
$\alpha_y(1) - \alpha_y(0)$ is akin to the Intention-to-Treat parameter in an RCT.

The estimator for τ is the Wald estimand:

$$\hat{\tau} = \frac{\hat{\alpha}_y(1) - \hat{\alpha}_y(0)}{\hat{\alpha}_d(1) - \hat{\alpha}_d(0)}$$

Fuzzy RDD: Parametric Regression model

Alternatively, one can consider the regression:

$$Y_i = a_y + X_i\beta_y + D_i\gamma_y + v_i, \text{ with } E(v_i|X_i) = 0$$

In this case $\hat{\tau} = \hat{\gamma}_y$

Fuzzy RDD: Local linear Regression model

Perform a local linear regression for both the outcome and the treatment indicator on both sides of discontinuity. For example, below the discontinuity:

$$\text{argmin}_{\alpha_{dl},\beta_{dl}} \sum_{i=1}^{n} \left(D_i - \alpha_{dl} - \beta_{dl}(X_i - c)\right)^2 I\{c - h \leq X_i < c\} \text{ and}$$
$$\text{argmin}_{\alpha_{yl},\beta_{yl}} \sum_{i=1}^{n} \left(Y_i - \alpha_{yl} - \beta_{yl}(X_i - c)\right)^2 I\{c - h \leq X_i < c\}.$$

The estimator for τ is the Wald estimand:

$$\hat{\tau} = \frac{\hat{\alpha}_{yr} - \hat{\alpha}_{yl}}{\hat{\alpha}_{dr} - \hat{\alpha}_{dl}}$$

Noack and Rothe [2021] propose new, easy to implement confidence sets (CS).

- They are always valid, but also at least as efficient as existent procedures.
- Construct CS based on inversion of the CS for the parameter $\tau_Y - c\tau_T$ for all $c \in \mathbb{R}$.

Confidence Sets in Fuzzy RD Designs

- Remember that the parameter of interest in fuzzy RD designs is: $\theta = \frac{\tau_Y}{\tau_T}$
- CS for θ are commonly based on its linear approximation and are therefore only valid for θ as long as approximation is ”good”.
- Noack and Rothe [2021] propose new, easy to implement confidence sets that do not rely on these approximations.
  - They are always valid, but also at least as efficient as existent procedures.
  - Construct CS based on inversion of the CS for the parameter $\tau_Y - c\tau_T$ for all $c \in \mathbb{R}$.

For replication

For replication:
`https://rpubs.com/phle/r_tutorial_regression_discontinuity_design`

A comparison of R-packages: Thoemmes, Liao, Jin (2016)
`https://journals.sagepub.com/doi/10.3102/1076998616680587`

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